

Kwangyoung Lee

HCI researcher, Ph.D Candidate

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Research statement

My research focuses on designing interactive systems that help individuals better understand and improve their mental health and well-being, drawing upon Human-Computer Interaction (HCI), Human-Centered Design, and Personal Informatics. I am particularly interested in how subtle and everyday human behaviors or analogies can reveal and externalize invisible personal states, such as emotional experiences and stress, into tangible and reflective data. Building on this, my work supports individuals in making sense of their personal data through structured, goal-oriented approaches and tailored interventions. Recently, I have expanded my research towards understanding how contextual and cultural constraints, such as workplace environments and social media influences, shape personal well-being practices. To address these constraints, I am exploring Multi-layered Personal Informatics, extending individual-level insights to broader ecological systems, and investigating multi-layered interactions between personal data, organizational contexts, and societal influences.

Education

Sep. 2022 -
Feb. 2026 (expected)

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D / Industrial Design

Thesis: Multi-layered Understanding of Personal Informatics: Exploring Contextualized Personal Data Across Intrapersonal, Interpersonal, and Institutional

Committee: Hwajung Hong (Chair), Taekjin Nam, Uichin Lee,
Takyoon Lee, Janghee Cho

Mar. 2016 -
Feb. 2018

Ulsan National Institute of Science and Technology (UNIST)

Professional Master of Design-Engineering

Thesis: Designing Toolkits for Self-Tracking and Self-Intervention to Improve Mental Health

Advisor: Professor Hwajung Hong

Feb. 2010 -
Feb. 2016

Korea Advanced Institute of Science and Technology (KAIST)

Bachelor of Science / Industrial Design

Research Experience

Aug. 2022 -
Present

Stress Management by Developing Digital Twins (@ KAIST)

Supported by the Institute of Information & communications Technology Planning & Evaluation (IITP)

Role: Project Manager, Research Assistant

- Developing human digital twins technologies for the prediction and management of emotional workers' mental health

- Conducting user research to explore how people perceive their emotions in the workplace with a smartphone application through data tracking and how they interact with data to relieve stress

Apr. 2018 -
Dec. 2019

Positive Computing through Persuasive Interactions (@ Seoul National University)

Supported by the Institute of Information & communications Technology Planning & Evaluation (IITP)

Role: Research Assistant

- Conducting a preliminary study on developing tools that can help manage stress by predicting and planning interventions
- Designing a calendar-based smartphone application for stress management and conducting user research to understand how people anticipate their daily stress based on the event, and how they mediate stress with an intervention

Feb. 2017 -
Sep. 2017

U-Glass Project (@ UNIST)

Role: Project Manager of the design team

- Developing auxiliary tools for an exhibition based on augmented reality (AR) technology
- Managing the design team to create a content scenario for experiencing museum exhibits with AR glasses and conducting user research to explore what will be an engagement factor in an immersive environment

Work Experience

Jan. 2020 -
Jul. 2022

Krafton (PUBG Studio), Seoul

Role: UX Designer

- Designing user experience (UX) for the lobby screen (outgame) of Battleground Game and constructing design systems for effective development
- Conducting design research to induce user retention

Jan. 2017 -
Feb. 2017

ITNJ (Start-up), Ulsan

Role: UX Research Assistant / Product Designer

- Developing online educational content and designing a portable tablet cradle
- Discovered new businesses based on the experiences of people using tablet PCs and designed the cradle to install the tablet

Jun. 2014 -
Dec. 2015

Coin-Cloud (Start-up), Daejeon

Role: CEO / Designer

- Developed a coin accumulation system using smartphones to reduce coin issuance cost
- Participated in a start-up competition hosted by KBD Industrial Bank and attracted investment from companies

Publication

DIS 2025
(Conditionally Accepted)

Thinking Outside the Data Box: Investigating the Potential of Data Manipulation for Self-Reflection on Personal Data

Yeohyun Jung, Kwangyoung Lee, Hwajung Hong

DIS '25: Proceedings of the 2025 Conference on Designing Interactive Systems

- CHI 2025*
(Accepted) **Peerserspective: A Study on Reciprocal Tracking for Self-awareness and Relational Insight**
Kwangyoung Lee, Yeohyun Jung, Gyuwon Jung, Xi Lu, Hwajung Hong
CHI '25: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems
- BigComp 2023*
(Workshop) **Designing a Personalized Stress Management System for Call Center Workers**
Kwangyoung Lee, Hyunseung Lim, Sooyeon Ahn, Taewan Kim, Hwajung Hong
2023 IEEE International Conference on Big Data and Smart Computing (BigComp)
- CHI 2020* **Toward future-centric personal informatics: Expecting stressful events and preparing personalized interventions in stress management**
Kwangyoung Lee, Hyewon Cho, Kobiljon Toshnazarov, Nematjon Narziev, So Young Rhim, Kyungsik Han, YoungTae Noh, Hwajung Hong
CHI '20: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems
- DESIGN 2018* **Aesthetic product interaction: The necessity of consistency between function and emotion**
K. Lee, J. A. Self, H. Hong
DS 92: Proceedings of the DESIGN 2018 15th International Design Conference
- CHI 2018* **MindNavigator: Exploring the stress and self-interventions for mental wellness**
Kwangyoung Lee, Hwajung Hong
CHI '18: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems
- DIS 2017* **Designing for self-tracking of emotion and experience with tangible modality**
Kwangyoung Lee, Hwajung Hong
DIS '17: Proceedings of the 2017 Conference on Designing Interactive Systems

Patent

- July, 2017* **Holder for embedding a tablet**
Role: Lead inventor
Patent No. KR2020170003972 (Expired)
- April, 2020* **Calendar-based intelligent stress management system and method**
Role: Lead inventor
Patent No. KR1024349770000 (Granted)
- August, 2023* **MindBattery provision system and method for improving emotion**
Role: Lead inventor
Patent No. KR1020230109004 (Patent Pending)

Teaching Experience

Teaching Assistant

Fall, 2016

Design Knowledge and Skills (Intro to Information Visualization)

Course: IID231, +20 students, UNIST

- An undergraduate course for teaching basic knowledge, tools, and practical skills for visualizing information with data.
- Provided feedback in the weekly one-on-one instruction session, guiding individual project development.

Spring, 2023

Needfinding Practice

Course: KEI430, +20 students, KAIST

- An undergraduate/graduate course for teaching design thinking based on observation of user behaviors.
- Provided feedback on the weekly group project development.

Award

April, 2018

Outstanding Presentation Award

ACM SIGCHI KOREA LOCAL CHAPTER

Academic service

Peer reviewing

ACM Human Factors in Computing Systems (CHI) 2024, 2025
ACM Designing Interactive Systems (DIS PWIP). 2025
Archive of Design Research (AODR). 2023, 2024, 2025